

DB-RLM: Bringing Recursive Language Models to Text-to-SQL

Final Presentation, Master Praktikum

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Why Text-to-SQL Is Hard

A Verifiable Task

- Maps natural language directly to SQL.
- Correctness is checked by **executing** the query, not by static reading.

The Static Schema Limit

- Small models lacking live data visibility fail predictably.
- **Structure:** Hallucinated columns and wrong join paths.
- **Values:** Filtering on nonexistent data or assumed formats.

Four Layers of Breakdown

1. Schema Selection:

Mapping entities to tables and finding valid join paths.

2. Semantic Reasoning:

Identifying grain, aggregations, denominators, and logic.

3. Execution Grounding:

Aligning model assumptions with what is *actually* stored.

4. Output Conventions:

Formatting the final output to match expected reference formats.

State of the Art: Recursive Language Models

The Paradigm Shift

- **Standard LLMs (In-Context):**
Stuff all data in context → context bloat & reasoning degradation.
- **RLM (Zhang et al., MIT CSAIL):**
Treats data as an **external environment** queried programmatically via code.

Three Core Mechanisms

1. **Programmatic Exploration:**
Inspect and filter data with code.
2. **Executable REPL:**
Maintains state and returns live feedback.
3. **Recursive Delegation:**
Sub-problem divide-and-conquer.

Reported Benchmark

RLM-Qwen3-8B outperforms base model by **+28.3%**.

Our Adaptation

Extend the RLM framework from text blobs to live, sandboxed SQL databases.

Research Questions & Experimental Setup

Research Questions

- **RQ1 (Inference Impact):**
How much can an inference-time DB-RLM system improve a frozen small model?
- **RQ2 (Ablation & Attribution):**
Which RLM mechanisms actually drive the accuracy gains?
- **RQ3 (Remaining Bottlenecks):**
What limits further scaling: reasoning, output conventions, or measurement?

Setup & Constraints

- **Generator:** gpt-5.4-mini (frozen, no fine-tuning).
- **Gold answers:** Offline scoring & post-run diagnosis only (no leakage).
- **Ablatable variables:** Environment, sandbox tools, schema context, and post-processing.

Core Experimental Design

Same model · same benchmark · same task
— only the system around the model changes.

Research Questions & Experimental Setup

RQ1

Improve a frozen small model at inference time?

RQ2

Which mechanisms actually matter?

RQ3

What still limits performance?

Same model · Same benchmark · Same task

→ Only the system around the model changes.

gpt-5.4-mini

| **No fine-tuning** |

No gold leakage

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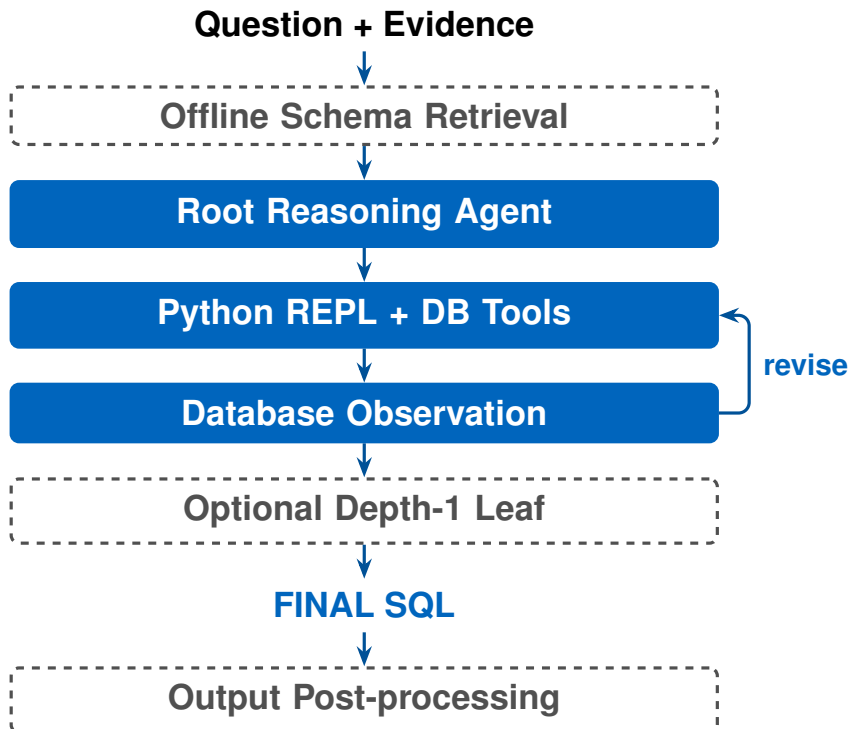
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DB-RLM Architecture



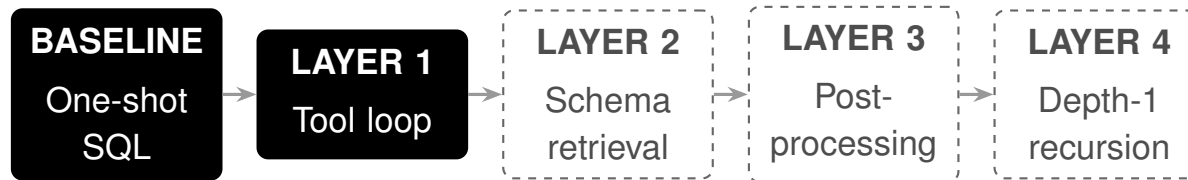
What the Agent Can Do

- `db.execute(sql)`
Run a query and observe the **real** execution result.
- `db.sample_values(table, col)`
Inspect what is **actually stored** to catch mismatches before filtering.

Loop Guardrails

- **Bounded Retry:** Repeats up to **8 turns** on SQL errors or empty result sets.
- **No Empty Finals:** Never finalizes on a query returning zero rows.

Architecture as Experimental Design



Add one capability at a time. Measure the change.

**Two targeted studies sit outside this chain:*

Negative control

Keyword-based schema pruning

Reasoning sweep

Low vs. high reasoning effort

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Use Case: From Hint to Correct Filter

Question: bird_1265

How many patients have a normal level of anti-ribonuclear protein and have been admitted to the hospital?

Hint

RNP IN (' - ' , ' + - ')

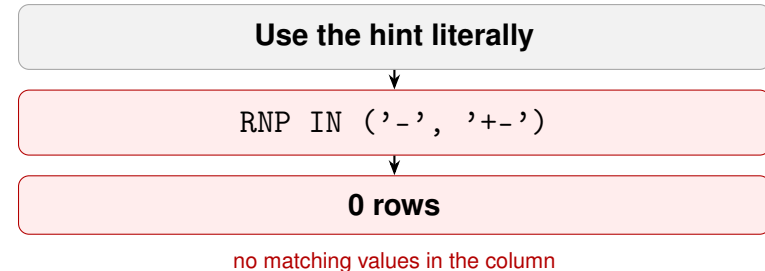
— → negative

+— → 0

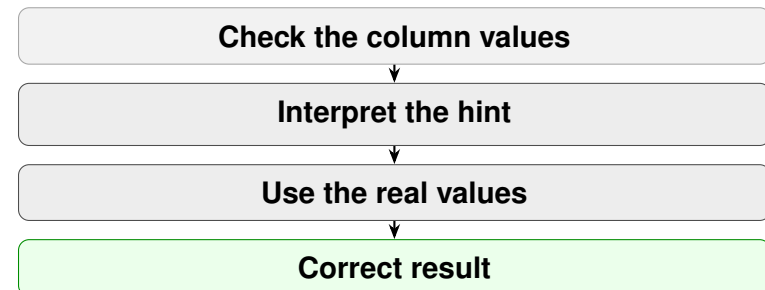
The problem

The hint uses symbols, but the database stores different values.

Without Database Access



With DB-RLM



The SQL is valid, but the filter uses values that are not in the database.

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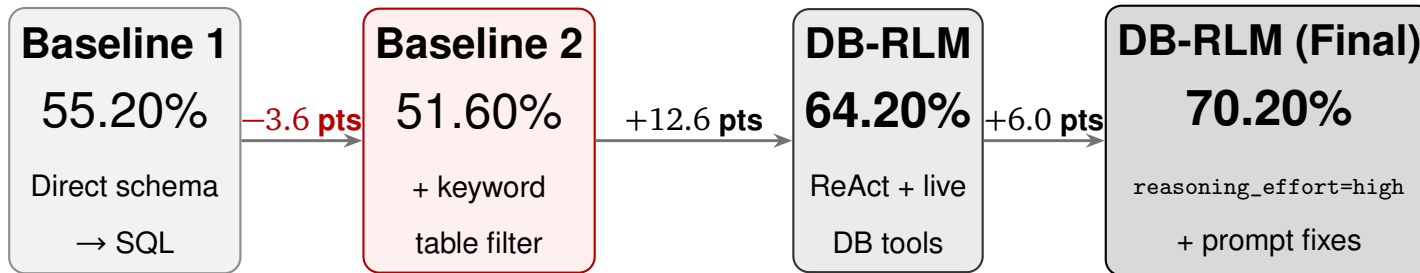
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Ablation Results



Latency / question: 1.5 s → 1.5 s → 3.4 s → 42.0 s
(LLM calls: 1 → 1 → 2.42 → 2.58. Token counts not instrumented in the earlier runs.)

Difficulty	Baseline 1	DB-RLM	△
Simple	71.6%	77.0%	+5.4
Moderate	50.8%	60.8%	+10.0
Challenging	42.2%	53.9%	+11.7

The tool loop helps most where it's needed most:
Live verification pays off more on hard questions, not less.

**The final +6.0 pt step bundles high reasoning effort (+2.2 pts in isolation), deduplication fixes, and removed hints.*

Reasoning Effort: Accuracy vs. Cost

Reasoning Effort	Execution Accuracy	Reasoning Tokens	Latency / question
low	68.00%	323	13.4 s
high	70.20%	4,419	66.8 s

+2.2 points — for $13.7\times$ the reasoning tokens and $5\times$ the latency.

Difficulty	low	high	Δ
Simple	79.1%	76.4%	−2.7
Moderate	65.2%	71.6%	+6.4
Challenging	58.8%	57.8%	−1.0

The gain sits entirely in the moderate tier.

More thinking helps find non-obvious workarounds, but on easy questions, it gives the model room to talk itself out of a correct answer.

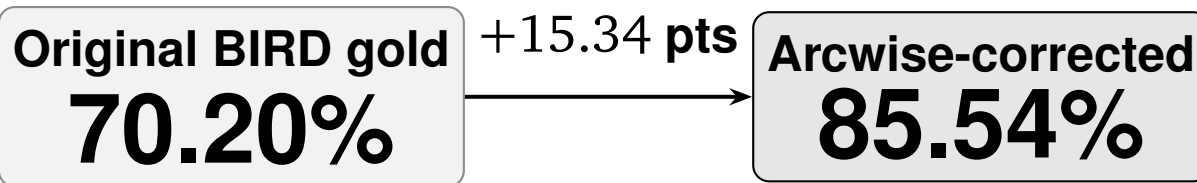
**Same system, same data — only reasoning effort changes. Full 500 questions, no timeouts either side.*

A Benchmark-Imposed Ceiling

Same model. Same system. Corrected benchmark.

Of 498 questions, the corrected version revised:

215 gold queries · **147** questions · **140** hints



31 of 64 classified failures (48%)

were traced to defects in the benchmark's own gold SQL or evidence hints.

Wrong numeric ordering

'93931' ranked above '177686' —
sorted as text, not as a number.

AND/OR precedence

Missing parentheses changed the query
result.

Defective evidence

The provided hint formula did not match
the task.

Model Scale Comparison

Same harness. Same configuration. Larger model.

Model	Effort	EX	Median reasoning tok	Calls	Median latency
gpt-5.4-mini	low	68.00%	209	2.44	10.2 s
gpt-5.4-mini	high	70.20%	2,949	2.58	42.0 s
gpt-5.4	low	71.60%	103	2.78	10.8 s
gpt-5.4	high	71.00%	1,683	2.90	32.4 s

Model size buys efficiency more than accuracy.

gpt-5.4 at low effort beats mini at high effort —
1.4 points better, with 28× fewer reasoning tokens.

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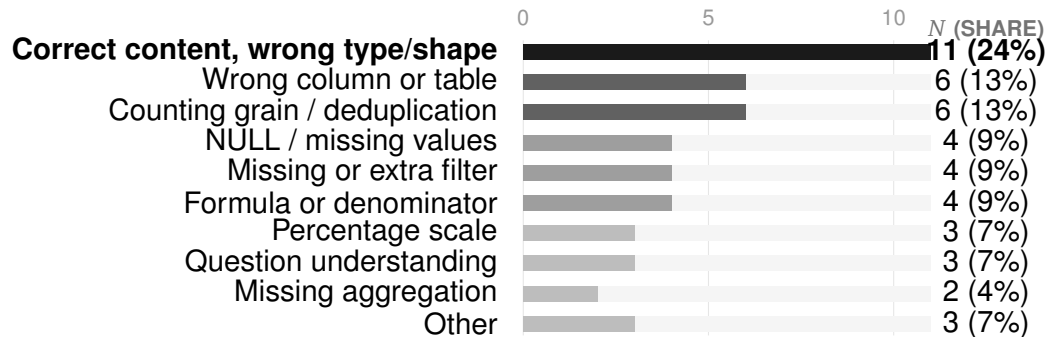
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What Are the 46 Remaining Errors?

46

inspected failures

Largest category: 11 cases (24%) the content is correct, but the returned type or shape is wrong.



Interpretation: the remaining errors span conventions, source selection, counting grain, missingness, filters and formulas.

Scope: this failure-selected diagnostic set is not a random sample; its shares are not benchmark-wide prevalence estimates.

How Do We Test a Claimed Error Cause?

Original trace 

$k = 1$ 

$k = 2$ 

$k = 3$ 

 fixed  resampled

**At every
checkpoint**

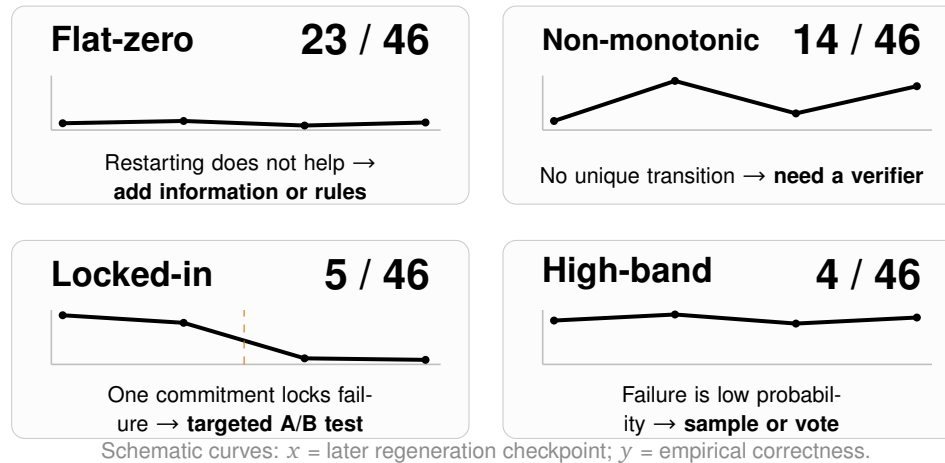
$N = 10$ independent
continuations
same model +
high reasoning

Accuracy at k
=
correct FINAL

valid continuations

We do not trust explanation text alone: we fix the prefix, replace the continuation, execute its tools, and measure the outcome.

Four Behaviors under Turn Resampling



Only **5 of 46** failures support a unique commitment point; “find the first wrong sentence” is usually not a sufficient diagnosis.

A Trace-Aligned Story Can Still Be Wrong

Case bird_637

“Which tags belong to Mark Meckes’s posts with no comments?”

Database fact

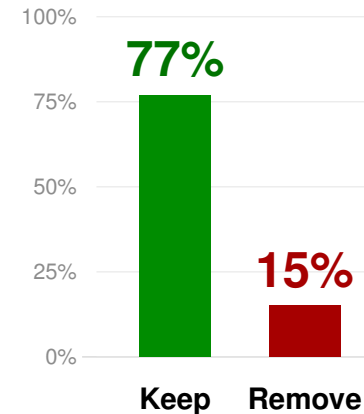
```
posts.Tags = '<books>';  
comments = 0
```

Observed error model
splits the stored string and
returns books

Plausible trace story “the
few-shot taught the wrong
tag structure”

A/B intervention: remove only the few-shot; keep everything else
fixed.

Correct continuations

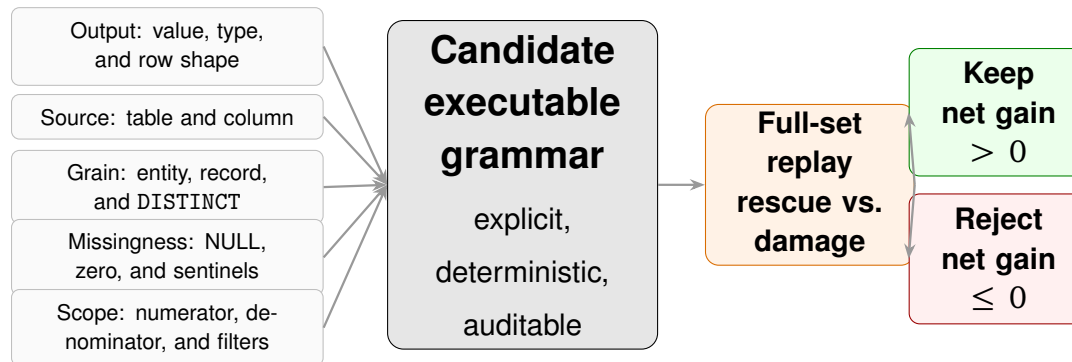


Hypothesis rejected

Keeping the few-shot gives 10/13 correct; removing it
gives only 2/13.

A trace suggests hypotheses. Only an intervention can test them.

From Recurring Decisions to Executable Grammar



Supported: recurring decisions exist; `keep_ties` and `printf_to_round` have positive full-set net value.

Not yet claimed: an automatically learned state machine or a grammar that generalizes to a second split.

Discovery proposes rules; counterfactual replay decides whether they ship.

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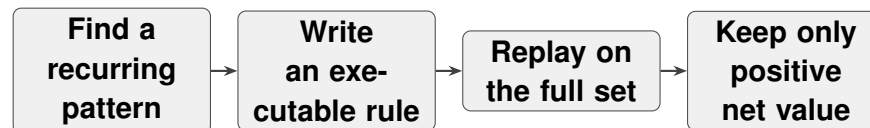
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What Did We Do after Finding the Errors?

Hypothesis from diagnosis	Test performed	Result and decision
The same model may fix the answer by trying again.	Resample every one of the 46 failures 10 times from multiple turn positions.	23/46 are Flat-zero; generic retry is not the main answer.
The few-shot may mislead bird_637.	Remove only the few-shot while keeping all other settings fixed.	Accuracy falls from 10/13 to 2/13; reject the explanation.
Tied rows should not be truncated to one output row.	Replay <code>keep_ties</code> bidirectionally on the full dataset.	Rescue 7, damage 2 in both runs; net +5.
Decimal <code>printf</code> formatting can be numeric rounding.	Replay <code>printf_to_round</code> bidirectionally on the full dataset.	Rescue 2–3, damage 0; net +2–3.



Generalization—and Why Not Just Route to a Larger Model?

Approach	What it solves	Role in this project
Large/small model routing	Select a stronger or cheaper model for each question.	A mature complementary approach; not tested here.
Reasoning distillation	Train a smaller model from stronger-model rationales.	A plausible future extension; not tested here.
Executable grammar	Compile recurring, measurable output differences into deterministic rules.	The framework developed in this project.

When can the framework generalize?

- Outputs can be compared structurally.
- Results can be scored or executed automatically.
- The same difference recurs across questions, ideally across databases.

Final takeaway

DB-RLM is the first executable, testable case. The broader goal is not to replace large models, but to convert recurring, measurable failures into reusable executable knowledge.

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